

SmartCare Health Monitoring with Smartwatches

Case Study Report: Business Data Analytics for Preventive Healthcare

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Executive Summary & Problem Identified

The Challenge: Aging Population in New Zealand

New Zealand's aging population (25% over 65 by 2040) strains the traditional reactive healthcare model. The SmartCare program leverages smartwatches for continuous, real-time monitoring of vital health metrics in vulnerable individuals over 50.

- ❏ **Key Finding:** 52% of patients are categorized as high or critical risk, primarily due to sleep deprivation and sedentary lifestyles.



52%

High/Critical Risk

Percentage of patients requiring urgent intervention.

82%

Health Score

Random Forest model prediction accuracy.

-0.70



Sleep-Stress Correlation

Strong inverse relationship (r) found.

Project Objectives & Business Value

This initiative converts raw smartwatch data into actionable clinical insights, supporting a shift from reactive to preventive care.

Identify Risk Factors

Find patterns and risk factors within the smartwatch data for early intervention.

Predictive Modeling

Develop models for weekly health scores and fall risks.

Patient Segmentation

Create patient groups based on risk for personalized treatment planning.

Decision Support

Develop an interactive Power BI dashboard for clinical decision-making.

Ethical Compliance

Ensure the process is ethically sound and protects patient privacy.



The value proposition includes minimizing hospital visits, enhancing quality of life through personalized plans, and optimizing resource use.



Data Analysis and Model Development

Methodology Overview

Data from 50 patients (age 50-79) over 4 weeks was collected, preprocessed (validation, outlier detection, standardization), and enriched with 8 engineered features (e.g., Age Groups, Risk Categories).

Analytical Techniques

- Exploratory Data Analysis (Correlations, ANOVA, t-tests).
- K-Means Clustering (K=4) for patient segmentation.

Predictive Modeling

Two Random Forest models were built using Python's scikit-learn library (70% train, 30% test).

Health Score Prediction

Random Forest Regression achieved $R^2 = 0.82$ (RMSE = 6.3).

Fall Risk Prediction

Random Forest Classification achieved **AUC-ROC = 0.81** (78% accuracy).

Feature importance analysis guided intervention priorities.



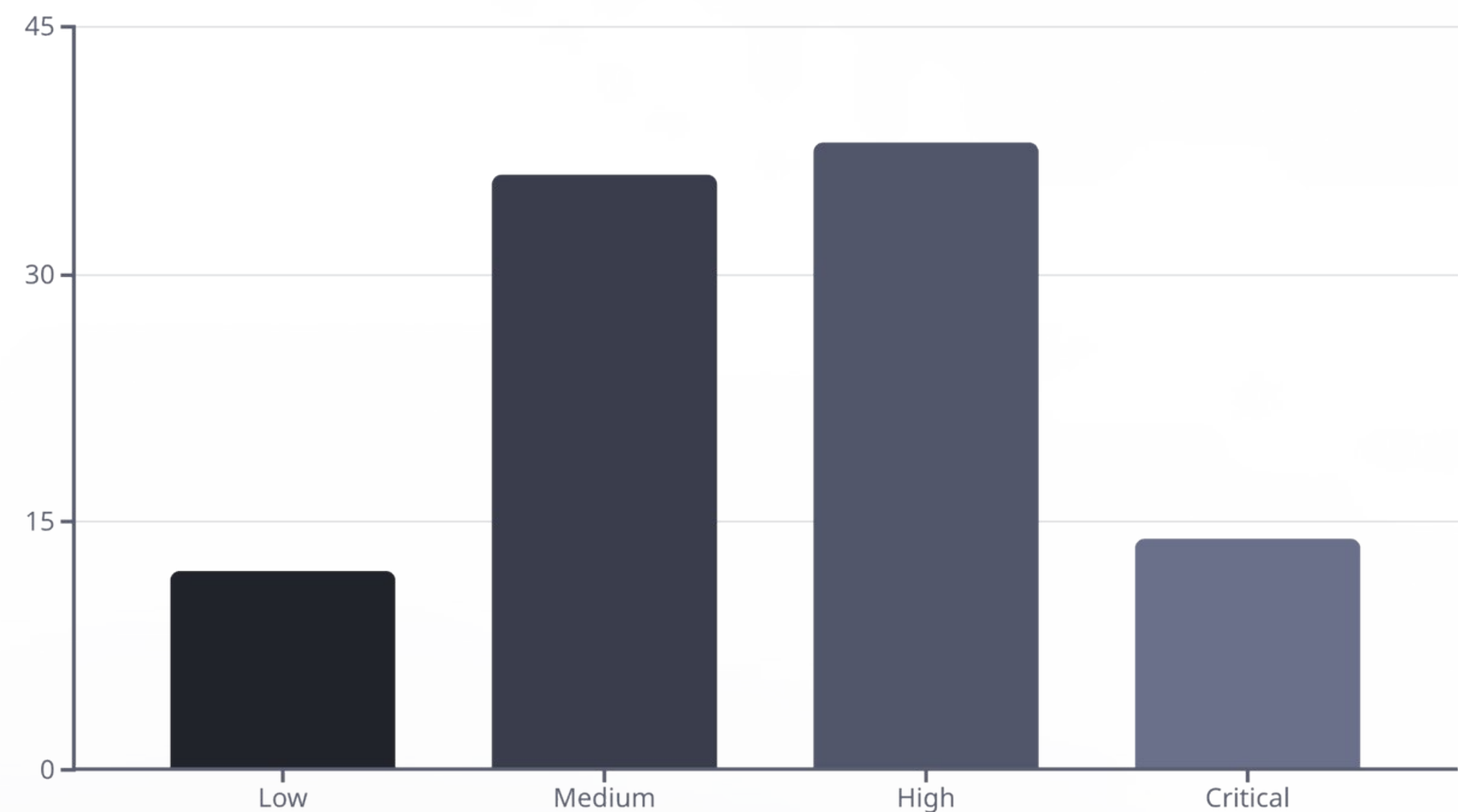
Key Findings: Health Metrics and Correlations

Demographic Profile (n=50)

- Mean Age: 67.4 years (50% are 70-79).
- Gender: 60% Female, 40% Male.
- Activity: 24% Sedentary, 52% Moderate, 24% Active.
- Stress: 30% of patients are above the high-stress cut-off (70).

Critical Correlation

The most robust association was between **sleep duration and stress index** ($r = -0.70$, $p < 0.001$). Each extra hour of sleep relates to an 8-point decrease in stress.



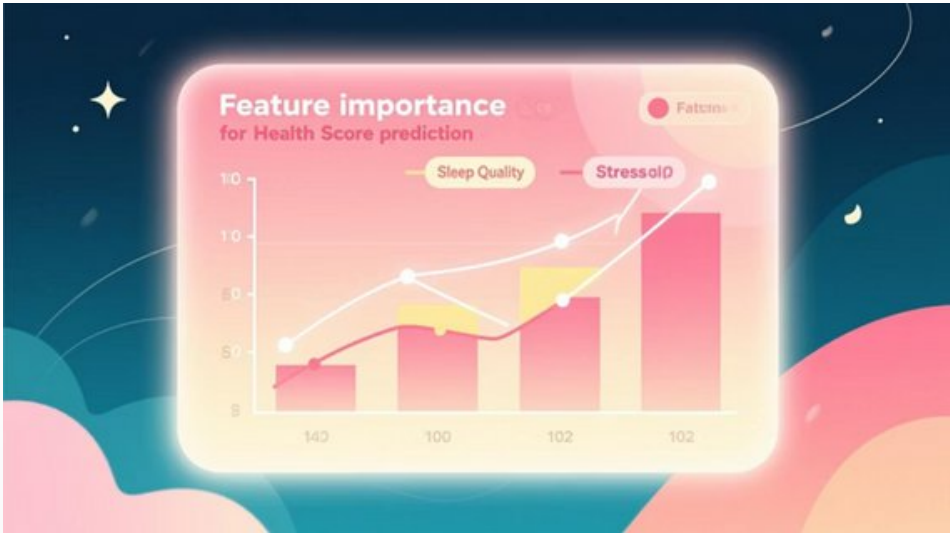
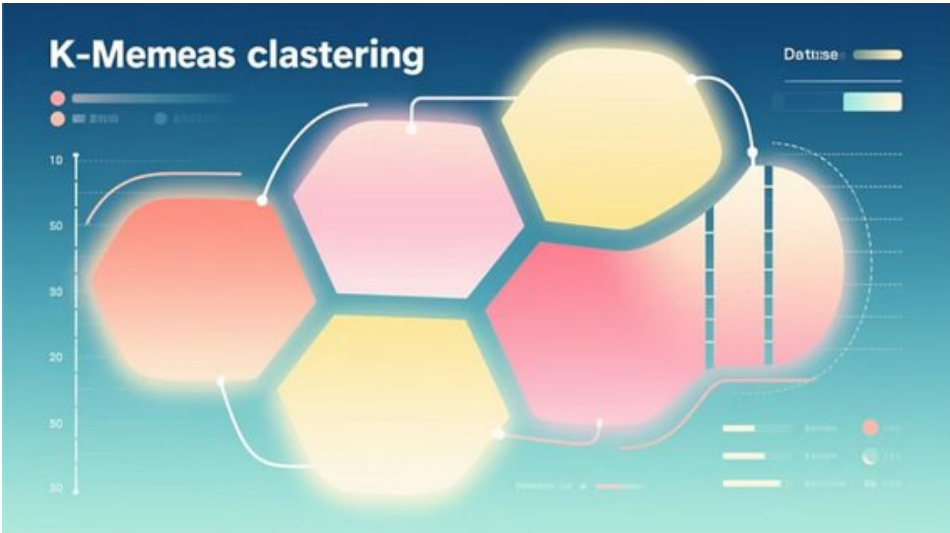
Risk Stratification shows 52% of patients are in the High or Critical risk categories.



Patient Segmentation and Predictive Insights

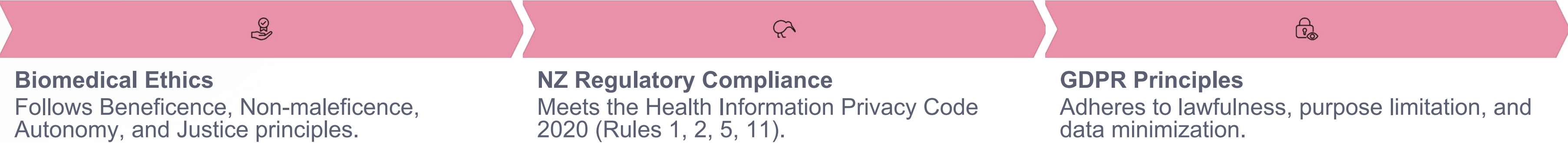
K-Means clustering identified four distinct patient profiles, guiding targeted interventions.

Cluster Profile	Patients (%)	Key Characteristic	Intervention Needs
Healthy & Active	18%	High Health Score (92.8), Low Stress (45.4)	Maintenance Support
High Risk - Intervention	34%	Low Steps (5,038), High Stress (69.7)	Intensive Multiple Interventions
Moderate Risk - Stress	20%	High Stress (78.0), Active Steps (9,160)	Psychological Interventions
Moderate Risk - General	28%	Average Metrics, Health Score (75.0)	Regular Monitoring



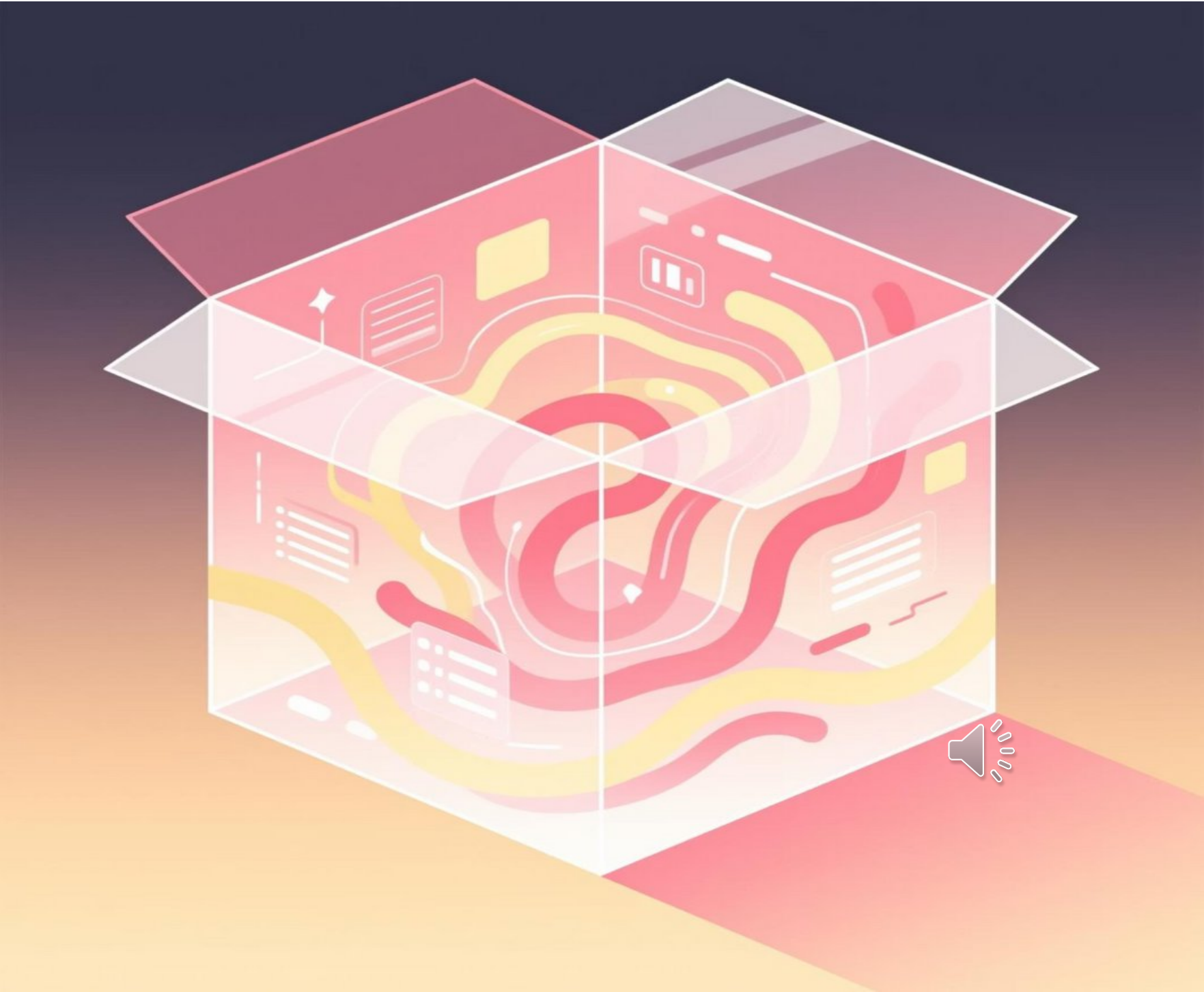
Ethical Framework and Regulatory Compliance

The project adheres to strict ethical and regulatory standards, prioritizing patient autonomy and data security.



Algorithmic Transparency

- Feature importance is disclosed to clinicians (e.g., Sleep Quality = 28%).
- Human-at-the-Loop (HAL) design requires clinical review before action.
- Bias audit confirmed no meaningful difference in model accuracy between genders or age groups.



Data Privacy and Security Statements

Privacy by Design is central to the SmartCare system, ensuring robust protection of sensitive health data.

Data Governance

Patient records are de-identified via pseudonymization. Data minimization excludes geolocation, financial, and communication data.

Data Security

Layered security: Role-Based Access Control (RBAC), multi-factor authentication, full audit trails, and quarterly privacy officer review.

Storage & Destruction

Data is processed on-premises (no cloud storage). Retention policy: program duration plus two years, followed by secure cryptographic erasure.

Transparency

Formal Privacy Impact Assessment (PIA) conducted. Patients can access their data and file complaints with guaranteed investigation within 30 days.



Priority Recommendations for Intervention

Interventions are prioritized based on feature importance and patient risk stratification.



Sleep Optimization

Target 24% of patients with poor sleep. Includes sleep hygiene training and specialist consultation. **Cost: \$2,400 (Pilot).**



Physical Activity

Target 24% sedentary patients. Implement progressive walking and strength training. Aim for 3,000 steps increase. **Cost: \$3,600.**



Stress Management

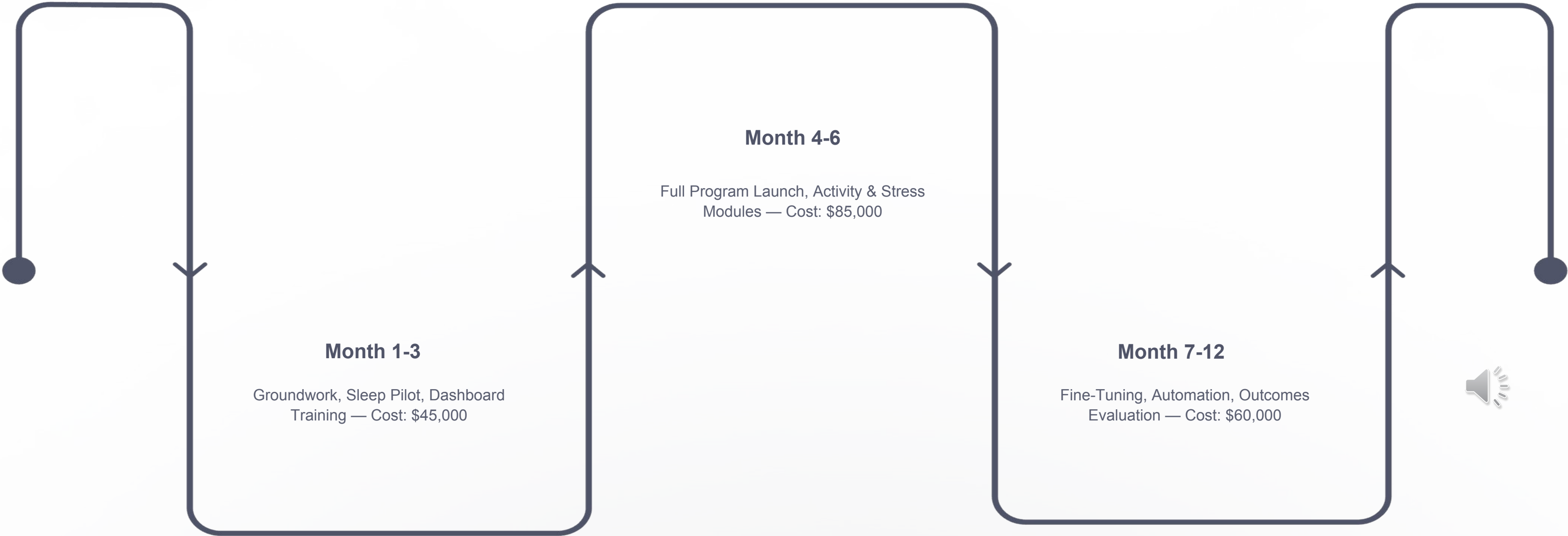
Target 30% of high-stress patients (especially Cluster 4). Includes mindfulness and therapy workshops. **Cost: \$6,000.**



Fall Prevention Protocol

Target 17 high-risk patients (age is the strongest predictor). Includes home safety workups and balance training. **Cost: \$13,600.**

Implementation Roadmap (Year 1)



Conclusion: Data-Driven Preventive Healthcare

The SmartCare project successfully demonstrates the power of smartwatch data to drive personalized, preventive healthcare interventions.

Key Takeaways

- Sleep, activity, and stress are the primary modifiable factors impacting health scores.
- Predictive models (81-82% accuracy) provide a robust early-warning system.
- The Power BI dashboard enables real-time clinical decision support.
- The proactive approach is both clinically sound and economically viable.

The full \$40,000 initial year outlay is anticipated to generate a **3:1 return on investment** within 18 months by preventing costly hospitalizations and ER visits.

